

7	(a)		$\frac{hc}{\lambda} = E_3 - E_2 \text{ (identifying which two levels.)}$ OR Uses wavelength of 658 nm $\frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{658 \times 10^{-9}} = E_3 - (-3.40 \times 1.60 \times 10^{-19})$ $E_3 = -2.42 \times 10^{-19} \text{ J}$	C1 (C1) C1 A1
	(b)	(i)	Energy of each photon, $E = \frac{hc}{\lambda} = 2.84 \times 10^{-19}$ Power = intensity $\times$ area $\left(\frac{N}{t}\right)E = \text{intensity} \times \text{area}$ $\left(\frac{N}{t}\right) = \frac{\text{intensity} \times \text{area}}{E}$ number per unit time, $\frac{N}{t} = \frac{(160)(2.5 \times 10^{-6})}{2.84 \times 10^{-19}}$ $= 1.4 \times 10^{15} \text{ s}^{-1}$	C1 M1 A0
		(ii)	Pressure = $\frac{\text{force}}{\text{area}}$ Pressure on the mirror by the light beam is equal to the total force of impact by the photons per unit area. Change of momentum for one photon after 'reflected' = $2p$ (where p = momentum of the photon) Total force exerted = Combined rate of change of momentum $= 2\left(\frac{N}{t}\right)p = 2\left(\frac{N}{t}\right)\left(\frac{h}{\lambda}\right)$ $= 2(1.4 \times 10^{15})\left(\frac{6.63 \times 10^{-34}}{7.0 \times 10^{-7}}\right) (= 2.65 \times 10^{-12})$ $\text{Pressure} = \frac{2.65 \times 10^{-12}}{2.5 \times 10^{-6}}$ $= 1.1 \times 10^{-6} \text{ Pa}$	C1 C1 A1

		(iii)	For the same intensity, replacing the beam of red light with <u>blue light that has a smaller wavelength will result in a decrease of $\frac{N}{t}$ ($\frac{N}{t}$ proportional to λ).</u> <u>Pressure is proportional to $\frac{N}{t}$ and inversely proportional to λ.</u> Thus the <u>pressure remains the same.</u>	M1 A1
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8	(a)	(i)	Let the power of the sun be P . Considering the sun as a point source, the energy	M1
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